



Original Report

Patient safety improvements in radiation treatment through 5 years of incident learning

Brenda G. Clark PhD ^{a,*}, Robert J. Brown RTT ^a, Jodi Ploquin MS ^b,
Peter Dunscombe PhD ^c

^aRadiation Medicine Program, The Ottawa Hospital, Ottawa, Ontario, Canada

^bRadiation Safety and Health Physics, The Ottawa Hospital, Ottawa, Ontario, Canada

^cDepartment of Medical Physics, Tom Baker Cancer Centre, Calgary, Alberta, Canada

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Abstract

Purpose: To quantify the impact of a comprehensive incident learning system in terms of safety improvements.

Methods and Materials: An incident learning system tailored for radiation treatment and based on published principles has been used consistently in our large academic cancer center for more than 5 years. In the adopted system, every incident, whether or not there is a resulting direct impact on a patient treatment, is recorded and investigated to determine basic causes. The scope of the program thus includes potential, or near miss, events which have no impact on patients but which provide valuable insights into program weaknesses and hence facilitate proactive measures to minimize risk.

Results: Analysis of 2506 incident reports generated over a 5-year period demonstrate a substantial decline in actual, nonminor incidents; ie, those with a dose variation from that prescribed of greater than 5%. Only 49 incidents (1.95%) had an impact on patients. The actual incident rate at the point of treatment delivery, the most vulnerable point in our process, has also decreased. The system has provided rapid feedback to monitor several initiatives including implementation of new technology and several new treatment techniques. Using the evidence provided by these incident reports, strategies were developed by a multidisciplinary team to address system weaknesses. Interventions introduced include several human error reduction strategies including forcing functions and constraints to improve system resilience.

Conclusions: Our results demonstrate that effective use of an incident learning system will strongly encourage the reporting of incidents, whether or not they directly impact a patient, and serve as a proactive means of enhancing safety and quality. As a side benefit, addressing and overcoming the cultural barriers between the 3 professional groups involved in radiation treatment has resulted in an improvement in the safety culture in our center.

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* Corresponding author. Radiation Medicine Program, The Ottawa Hospital, 501 Smyth Rd, Box #927, Ottawa, ON K1H 8L6, Canada.

E-mail address: brclark@toh.on.ca (B.G. Clark).

Introduction

The types of errors encountered in radiation treatment have been described extensively and include a wide range of systematic and random errors,¹⁻⁵ the implementation of new technology,⁶⁻⁸ and stressors in the clinical environment.⁹ Recent work has described several methods to improve the safety of radiation treatment^{10,11} and the basic requirement for many of these is an effective means to analyze and identify risk, inherently relying on the reporting and follow-up of actual and potential errors and incidents.¹²⁻¹⁶ We report results generated over a 5-year period from the use of a comprehensive incident learning system in a large academic radiation treatment program.

Methods and materials

Our center is a comprehensive clinical and academic cancer center delivering approximately 4000 radiation treatment courses on 12 megavoltage units annually; 9 conventional accelerators, 2 tomotherapy units, and a CyberKnife (Accuray, Sunnyvale, CA). An incident in our system is defined as “an unwanted or unexpected change from a normal system behavior that causes or has the potential to cause an adverse effect to persons or equipment.” Every incident, whether or not there is a resulting direct impact on a patient treatment, is recorded and investigated to determine basic causes and to identify corrective actions to address system weaknesses. During selection of an appropriate intervention, a conscious effort is made to select relatively stronger actions according to the suggestions given by the US National Center for Patient Safety.¹⁷ Examples of stronger actions are architectural or physical changes, new devices and interlocks, and process simplification. Examples of weaker actions are double checks, warnings, individual reminders, new procedures, or training. Information selected and used is specific to each incident.

Regular feedback is provided to the program staff emphasizing improvements that have been made as a result of follow-up of an incident report, a reinforcement of the continuing importance of the program and the fundamental change in the incident handling process. Incidents of high educational value are identified and are included as case studies, reviewed with the Radiation Medicine Program when quarterly statistics are presented. More recently, we have recognized individuals with the presentation of a series of coffee cards awarded to the individuals logging the most significant reports. Careful attention is made to assign corrective actions appropriately, recording the completion of the corrective action.

The system has been described elsewhere^{18,19}; operational details are given in the supplementary materials (available online only at www.practicalradonc.org).

Table 1 Incident statistics for calendar years 2007-2011

Variable	2007	2008	2009	2010	2011
Incident reports	658	686	461	416	285
Minor/potential (%)	96.2	99.0	98.0	99.5	97.9
Actual	117	84	62	51	54
Actual, nonminor	25	7	9	2	6
Actual treatment unit (%)	0.10	0.04	0.03	0.02	0.03
Reporting index ^a	25	97	50	207	47

^a Ratio of minor or potential incidents to nonminor actual incidents.

Initially, paper documentation was used exclusively for reporting, investigation, and follow-up of incidents with a spreadsheet used to summarize and track incidents, monitor follow-up, and to provide data for statistical analysis. During the last year, the reporting part of the system has been migrated to a commercial software package configured to mimic the paper system.

Results

The most significant result of this project has been the change in culture to one where hundreds of incidents are freely reported per year without fear of reprisal. Prior to the system implementation, radiation-related incident reporting in our center was limited to serious and critical actual incidents with little or no focused investigative follow-up. Levels of compliance varied greatly by year and work area. Incident reporting was largely viewed as a possible prelude to disciplinary action. There was no root cause analysis done and very little feedback or learning from incidents. There was no structured feedback to the staff, reports were

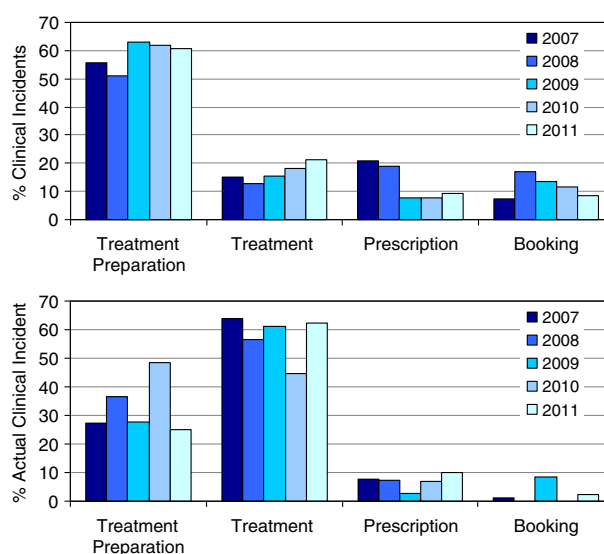


Figure 1 Distribution of clinical incident origin for all incidents (top) and actual incidents only (bottom).

passed between successive quality assurance committee chairs and, at one time, were even lost. Overall, the entire process was chaotic and unproductive. With the incident learning system, the staff appear engaged and empowered and, although hard to measure, we have seen improved staff morale as employees now feel that they can make a difference in the safety and quality of patient care.

Our quantitative analysis is derived from a total of 2506 incidents reported during calendar years 2007-2011; statistics are given in Table 1 and work area origin

illustrated in Fig 1. During this period, a total of 345,792 individual treatments were delivered, giving an incident rate of 0.7%. Figure 2 shows the distribution of actual incidents reported at the treatment unit over the 5-year period and Fig 3 quantifies the incidents originating during treatment planning. The primary basic causes assigned after investigation are summarized in Table 2. Incidents classified as nonminor are where a dose variation from prescription of greater than 5% is detected or a near miss or unsafe condition occurs which could potentially cause a treatment error. (Full details of the severity scale are provided in the supplementary material, available online only at www.practicalradonc.org.) Details of the 49 actual nonminor incidents are given in Table 3, a list of the major interventions introduced is given in Table 4.

Discussion

A true metric for measuring safety in radiation oncology is difficult to identify; we do not have outcomes such as a reduction in mortality that can be clearly tied to our interventions. We have used incident reporting here as a surrogate to indicate changes in our program safety but this must be identified as an indirect measure of system safety. It is clear from the data (Table 1) that the number of actual incidents reported has dropped during the 5 years and also that the severity of actual incidents has fallen. We attribute the reduction in nonminor actual incidents to the increasing awareness of patient safety issues, together with interventions and process changes we have introduced as a result of incident investigation and analysis. This reduction is particularly significant given that several major changes in process and technology have occurred in the same time period, resulting in an increase in our treatment capacity of 20% and in staffing of nearly 30%. The support of the senior management (departmental chiefs) has meant that quality improvements arising from incident findings have been instituted rapidly. The collaborative interprofessional approach has facilitated identification and resolution of issues in a timely fashion. In the first year alone there was a 54% decrease in serious incidents and this is a trend which has continued. This approach has also made supervisory staff more aware of what are weaker and stronger actions enhancing the effectiveness of interventions.

However, the number of reports has also fallen, especially during 2011 (Fig 4). While we are reasonably confident that all nonminor actual incidents are reported due to the very strong culture of professional accountability in our center, we recognize that voluntary reporting systems are unlikely to capture all events and there will certainly be variability in the reporting of minor and potential incidents. However, a reduction in incident reporting may simply reflect apathy rather than a true reduction in incidents. One important factor in this

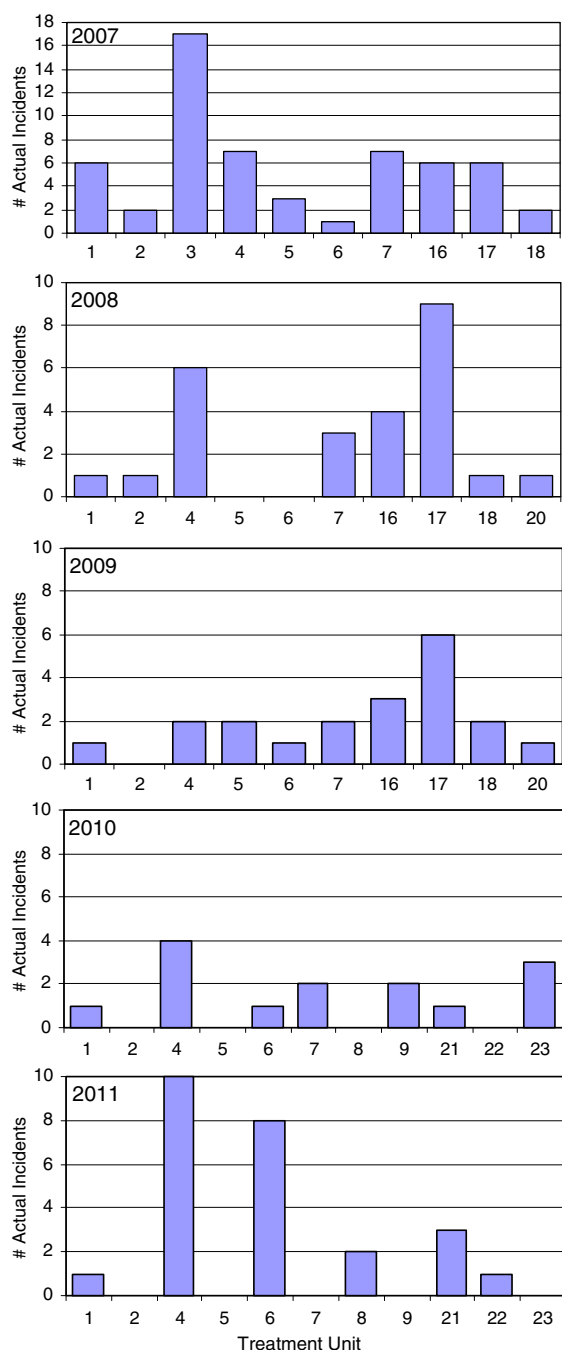


Figure 2 Actual incidents originating at the treatment unit. The vertical scale of the top plot is larger than the other plots.

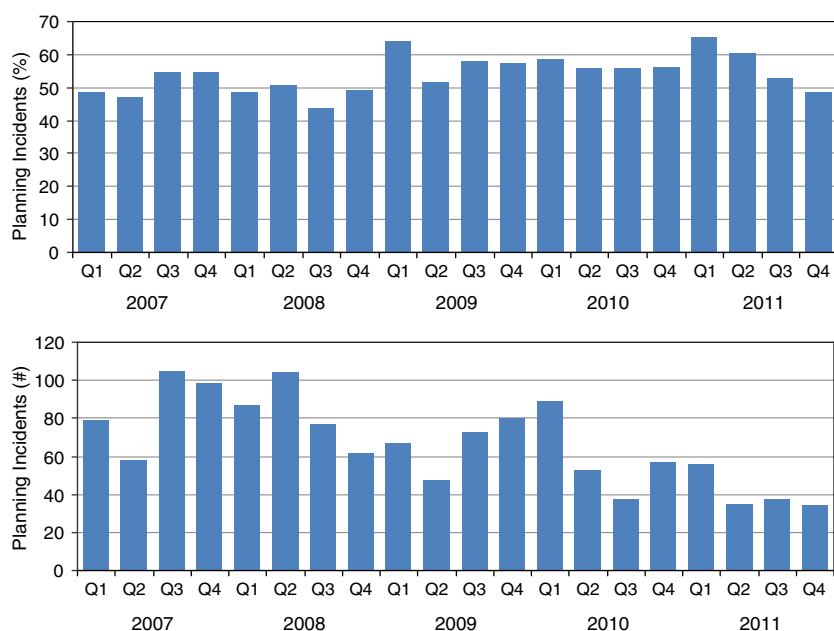


Figure 3 Incidents originating in treatment preparation (planning) as a percentage of the total number of incidents reported (top) and in absolute terms (bottom).

reduction is the implementation in 2010 of a hospital-wide electronic reporting system. Despite considerable efforts to customize the reporting interface, the use of this system is nonintuitive and cumbersome. After implementation of this system, the quarterly reporting rate fell from an average of 150.8 reports to 78.6. In an effort to restore reporting rate, the option of using a paper form for reporting has temporarily been restored while a custom online solution is developed.

Analysis shows that a consistent 70%-80% of incidents are primarily caused by either lack of, inadequate, or failure to follow standard procedures (Table 2), which is a result also found by other authors.^{20,21} The second most common basic cause initially was communication issues (unclear, inadequate, misunderstood, conflicting) but more recently, work planning issues (inadequate, conflicting priorities, personnel availability) have been identified.

Table 2 Percentage of incidents assigned to each basic cause category

Basic cause classification	2007	2008	2009	2010	2011
1 Standards/procedures	81.8	76.5	79.0	66.8	69.1
2 Materials/tools/equipment	1.2	3.1	1.5	3.9	2.8
3 Design	2.0	0.9	0.4	1.9	2.1
4 Work planning	2.9	3.9	8.0	11.3	11.6
5 Communication	7.6	9.6	6.9	9.4	8.1
6 Knowledge/skill	3.2	3.6	3.3	5.0	2.1
7 Personal capabilities	0.6	0.2	0.2	1.2	3.2
8 Personal judgment	0.3	1.9	0.7	0.2	1.1
9 Natural factors	0.3	0.3	—	0.2	—

The distribution of incident origin reveals that although the treatment preparation work area was the source of consistently more than 50% of all incidents reported, in terms of actual incidents having an impact on patient treatments, the treatment delivery work area was the more vulnerable (Fig 1). This is most likely a reflection of the fact that while most of the treatment parameters are generated automatically in the treatment planning software and transferred electronically to the treatment delivery unit, the precise setup and placement of the patient relies on instructions manually transcribed into the information system. This is a recognized system weakness and strategies are under consideration to address this issue.

Since 2007, our numbers have stabilized somewhat; we have made progress with reducing some types of errors but others have proved more intransigent. For example, we have seen a reduction in geographic miss incidents but those resulting from incorrect positioning in the beam have not been reduced. Also, bolus errors were reduced for several years but during 2011, a smaller, more complex-shaped bolus was introduced and this has proven challenging. Of the 10 bolus errors reported in 2011, 4 were omission of bolus from treatment, 2 were manufactured incorrectly, and the remaining incidents were poor or incorrect positioning. Of note, the use of bolus is a manual intervention relatively more difficult to control.

Of concern, 16 of the 49 (35%) nonminor actual incidents (ie, those impacting a patient with a dose difference greater than 5% from that prescribed) involved incorrect shifts of the patient during alignment prior to treatment (Table 3), a finding recently reported elsewhere.²² However, the rate of actual incidents occurring at the treatment unit decreased from 0.10 incidents per treatment

Table 3 Details of the 49 actual nonminor incidents**Critical**

1. Shifts incorrect from CT marks, wrong volume treated to 12 Gy
2. Wrong area in lung targeted, planned, and treated

Major

1. Shifts incorrect from CT marks, geographic miss
2. Shifts incorrect, from wrong tattoo, wrong volume treated on 1 day
3. Incorrect HDR brachytherapy source placement, geographic miss
4. Asymmetric field size incorrect, 4 treatments
5. Error in beam data for dose calculations (Operational)
6. Collision of robot with cart during measurements, \$20,000 repair and 4.5 hours of downtime (Operational)
7. Planning system file corruption, prostate implants rescheduled (~\$2800 to replace seeds lost) (Operational)
8. Radioactive seed shipment within hospital (Operational)

Serious

1. Shifts incorrect from CT marks or tattoos (13 incidents)
2. Pacemaker issues (2 incidents)
3. Treatment canceled, oncologist did not inform treatment unit
4. Concurrent chemotherapy scheduling issue
5. Wrong shielding used, 5 treatments
6. Wrong patient treated
7. Confusion regarding treatment volume
8. Patient canceled, treatment on call, record lost, patient restarted, canceled by resident
9. Field exit through chin
10. Manual calculation error, dose doubled, 1 treatment of 10
11. Targets completed but not appended, patient delayed
12. Eye shielding incorrect, 3 treatments delivered
13. Planning system shutdown, 1/2 day lost (Operational)
14. Blood work requisition misassigned (Operational)
15. Cobalt source would not deploy, partial treatment (Operational)
16. Isocentric couch orientation reversed in console software (Operational)
17. Routine imaging quality assurance tests incomplete (Operational)
18. Door interlock/breaker switch malfunction (Operational)
19. Quality assurance device dropped, \$2500 (Operational)
20. Hardware failure during treatment, no physicist available (Operational)
21. Micro-multileaf collimator cover damaged during collision test (Operational)
22. Treatment accessory dropped on coworker (Occupational)
23. Chemical spill, splashed into eyes/mouth (Occupational)
24. Aggressive patient (2 incidents) (Security/Other)

Table 4 Summary of major interventions implemented**General**

- Changes in staffing levels to concentrate effort in vulnerable parts of the process
- Modification of the quality assurance processes to focus on weaknesses in the treatment preparation process
- Implementation of chart and film audits
- Increase in peer review of target contours and treatment plans
- Identification to the manufacturer of the potential for error incorporated within separate software systems; ie, 3 communicating treatment planning software systems using 3 different geometric coordinate systems
- Initiation of a failure mode and effect analysis of out-of-hours emergency treatment planning and delivery process

Manual intervention improvement

- Reduction of transcription tasks, including several achieved through software upgrades
 - Physical plant changes such as barriers limiting patient access to controlled areas
 - Constraints and forcing functions such as the use of checklists requiring sequential completion of quality assurance through the process from planning to delivery
 - Decommissioning of techniques performed rarely
 - Minimizing reliance on memory through additional visual aids, additional in-room monitors
 - Template standardization in several areas
 - Treatment table indexing
 - First day treatment checklists
 - Where possible, treatment units are scheduled to be site specific
 - Treatment delivery work flow streamlining
 - Increased staff training
 - Addition of cognitive aids where appropriate
 - Restriction of the use of 2 technologies identified as being more likely to result in error
 - Root cause analysis process amended to focus on physical, cognitive, and organizational factors
 - Inclusion of multifactorial analysis where required
- Policy and procedure changes**
- Verbalization of treatment parameters prior to treatment delivery
 - New laterality policy
 - Modifications to the pacemaker policy
 - Use of physical measurement instead of volume imaging for simulation
 - Standardization of beam naming across planning system, calculation data, and information system
 - Documentation of overrides
 - More rigorous policies on staff accreditation for emergency treatments delivered outside clinical hours
 - Adjustment of treatment delivery quality assurance process
 - Improvement in reference dosimetry, increase in redundancy, and standardized template
 - Increase in redundancy for critical procedures: double checking of physics work, independent checking of commissioning work

fraction delivered in 2007 (Table 1; Fig 2). The data from 2007 accelerated the retirement of our oldest treatment unit (unit 3, a cobalt unit) by providing evidence of increased risk with 18 incidents in 2007, 2.6 times the average for all

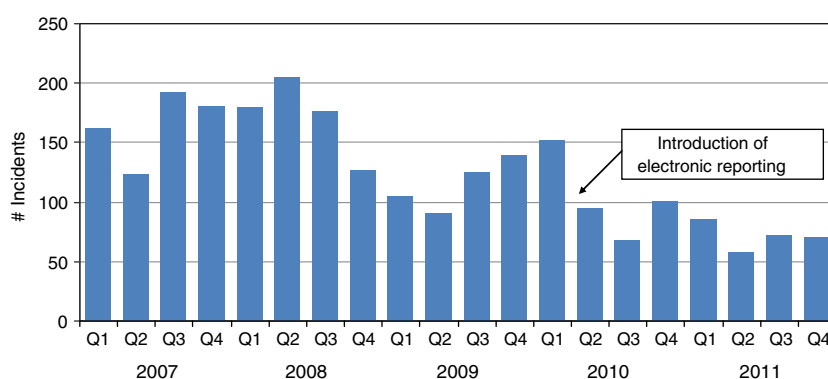


Figure 4 Number of incident reports generated per quarter.

units in that year. The data from 2008 and 2009 indicated the risk inherent in a treatment unit with no electronic imaging capacity with the only remaining unit not thus equipped, unit 17, recording 3.4 times the average for that year. This result supplied the evidence to justify the \$500,000 per unit of adding on-line imaging capability. Of note from the plots for 2011 is the relatively high number of incidents recorded on units 4 and 6. Unit 4 has a high palliative workload, where a large number of patients are treated with manually entered treatment parameters, and 5 of the reports from unit 6 related to bolus errors.

Finally, these data highlight the advantages of an integrated system. Over the 5-year period, we recorded 142 actual incidents originating at the treatment units, only 7 (5%) of which occurred on a tomotherapy unit. Additionally, of the 49 actual nonminor incidents, 27 (55%) originated at a treatment unit but none on a tomotherapy unit. The tomotherapy treatment unit has integrated treatment planning and imaging systems, leading to minimal data transfer, intrinsically providing a safer work environment. Additionally, in our center the radiation therapy staff on our tomotherapy units also perform all treatment planning for their patients, increasing their technical understanding of the treatments and eliminating staffing hand-offs between treatment planning and delivery.

Incidents originating in treatment preparation remained consistently at an average of approximately 53% of the total. However, when plotted as absolute numbers, a rise in the second half of 2007 is apparent, coinciding with the introduction of new treatment planning software and a new paperless and filmless procedure (Fig 3). Despite careful preparation, this implementation gave rise to an increase in minor incidents reported, the investigation and analysis of which proved to be a very useful means to identify and resolve issues. The rise in 2009 prompted the development of an enhanced quality assurance process and the lower rate of reports has been maintained since implementation of this process in 2010.

In summary, there were several general messages in these data and they are listed below. While some of these are obvious, we have found that having the data to support the actions identified to address these issues has been

invaluable, particularly where there is a cost associated with remediation.

- (1) Thirty-five percent of the highest severity (non-minor) incidents reported were caused by incorrect patient setup, one of the manual tasks.
- (2) The majority of actual incidents arise from manual intervention of one type or another.
- (3) In our experience, the tomotherapy treatment units are noticeably less error prone with only 7 incidents recorded on these units during the 5-year period, none of which impacted patients.
- (4) In our experience, some treatment units are more error prone (old technology, units that treat a wide variety of sites versus site-specific units, units with staff that rotate regularly versus other units with a specialized team).
- (5) The treatment of patients outside clinical hours under emergency conditions gave rise to an estimated 10% of nonminor incidents over this period. This process is particularly problematic as full resources are not available and a simplified treatment planning process is used which is not fully automated. As a result of these data, a failure modes and effect analysis (FMEA) has been initiated to review this process and propose improvements.
- (6) The system has provided an important monitoring and feedback process during implementation of new technology and software.

A final asset derived through the use of this system is the breakdown, or partial removal, of the barriers among the 3 main professional groups in radiation treatment. This type of barrier is a well recognized challenge in health care, where multidisciplinary teamwork is common, and has been identified as having a high potential for increased risk. With our implementation of this system, we have made headway in bridging the professional silos between radiation therapy, radiation oncology, and medical physics through a shared focus to improve the safety and quality of radiation therapy in our program. The challenge of engaging physicians in a reporting culture is a well-

known barrier in health care and requires more than removing blame or establishing reporting processes.²³ Our system has stimulated medical ownership in the processes of reporting and enhanced awareness of the function of reporting by including radiation oncologists in incident investigation, causal analysis, and follow-up for relevant reports. We have found that implementation of changes designed and brought forward by “one of their own” has been accepted within each professional group, including the physicians. The added motivation provided by the quarterly awarding of \$10 coffee cards has also helped! Representatives from every professional group have presented the work of our committee at the meetings of their respective professional organizations, reinforcing ownership and pride in our achievements.

Conclusions

In the absence of an ideal metric to measure patient safety, incident reporting may be our best surrogate and we have presented quantitative evidence that implementation of a structured incident learning system customized to radiation treatment has increased safety in our program and reduced errors impacting patients. Feedback is provided to all stakeholders on a regular basis and staff satisfaction has been evident by improved morale. One of the major benefits has been the building and strengthening of multidisciplinary partnerships, increasing engagement of staff and physicians, and providing an enabling environment to promote and improve patient care through the application of quality improvement initiatives developed as a result of direct staff feedback. We believe we have created a safety culture where every team member's opinion is valued, regardless of position or seniority. The radiation treatment process, as with other medical interventions, is vulnerable to human error and we have implemented several interventions aimed to reduce these types of errors, including reduction of transcription tasks, standardization, table indexing, and additional policy and procedure changes addressing verbalization of treatment parameters prior to treatment delivery, laterality, and documentation of overrides. Some of these interventions are not particularly innovative; however, the transparency and multidisciplinary support generated by our system has increased the number and success rate of these program initiatives.

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